

# Conner Moulaison

**Phone:** (360) 961-5549   **Email:** hello@connermoulaison.com   **Portfolio:** connermoulaison.com

Game programming generalist experienced with low-level engine development and a broad range of game systems

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## Skills

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| <b>Networking:</b>    | Client-side prediction/server reconciliation supporting both dedicated and player-hosted servers                                                           |
| <b>Gameplay/Math:</b> | Physics, including 3d movement and collision resolution, gameplay systems including economic simulations, spatial voxel puzzles, and arcade platformers    |
| <b>Build/Tooling:</b> | 3d level editing tools, in-game debug visualizations, asset packing pipelines                                                                              |
| <b>Performance:</b>   | Custom memory allocators, profiling and optimization, cache-friendly code                                                                                  |
| <b>Graphics:</b>      | Vulkan/OpenGL, text rendering, 3d textured meshes with Blinn-Phong lighting, compute shaders                                                               |
| <b>Collaboration:</b> | Gathering requirements and executing large, interdepartmental service rollouts, creativity and problem solving, clear communication, persistent work ethic |

## Solo Projects

**Ceres:** Asteroids inspired 3d shooter prototype built in a custom data-driven engine | *C, OpenGL, Xlib, ALSA*

- Data driven codebase with custom editor tools, an automated asset packing pipeline, and C code generation
- Game layer built separately and dynamically linked at runtime, allowing hot reloading of game logic
- In-engine level editor which can be dropped in and out of in the middle of gameplay
- Clean, readable code in a mixture of imperative and data-driven styles and a simple, cache-friendly memory model

**Netpong:** Online multiplayer implementation of Pong | *C++, OpenGL, Xlib, Unix Sockets*

- Client-side prediction and server reconciliation supporting both standalone and player hosted servers
- Custom UDP transport layer driven by Unix sockets with session management and bit packing for network packets
- Debug tools for simulating varying network conditions, including latency and packet loss

**Crumble King:** Arcade platformer released on itch.io for Windows and Linux | *Odin Language, SDL 2, MiniAudio*

- Complete, polished arcade platformer featuring 6 levels, 3 playable characters, and unlockable game variants
- Live synthesized audio system emulating "Tandy sound" chip driven by a custom music track format

**Cicero:** Compiled programming language targeting x86-64 assembly | *C, NASM Assembler*

- One-pass recursive descent parser to emit x86-64 NASM assembly and display error messages, implementing basic constructs such as variables, operator precedence, conditionals, and jump instructions

## Professional Work

**System Administrator:** June 2023 to March 2025 | *Barleans Organic Oils, LLC*

- Led the direction of the department and built documentation and standards, hiring and leading a help desk specialist
- Implemented improvements in SQL, Excel, JavaScript, Python, Powershell, and VBA for data reporting/modification, mass customer communication, change control, and helpdesk support
- Made more than 100 changes to our Shopify front-end in HTML/CSS/Javascript to adhere to accessibility standards, including modifying modal logic within various page elements
- Introduced custom fields and behaviors in our Sage X3 ERP system using both SQL and Sage's proprietary language, with changes to our financial and promotional data import processes, for example, saving 5-10 hours of work per week